

AWAIS SHAH

(+92) 306-5141475 • awais030380@gmail.com • Haripur, Pakistan
GitHub: github.com/AwaisShah75 • ORCID: [0009-0007-0928-0378](https://orcid.org/0009-0007-0928-0378)

Computer Science graduate with one published and one under-review research paper in real-time object detection for resource-constrained embedded systems. Experienced in designing and deploying lightweight vision models (SSD MobileNet V2, YOLOv8) with a focus on performance–efficiency trade-offs in CPU-based and edge environments. Proficient in Python-based ML workflows including TensorFlow, OpenCV, and Matplotlib. Seeking graduate research opportunities in Efficient AI, Edge Intelligence, and Embedded Computer Vision.

EDUCATION

B.Sc. Computer Science — University of Haripur, Pakistan — 2020–2024 — CGPA: 3.14/4.00

PUBLICATIONS

A Review of Real-Time Deep Learning–Based Object Detection Models for Resource-Constrained Embedded Systems

NUML International Journal of Engineering and Computing (NIJEC), 4(2), 13–29. (2025) • DOI: [10.52015/nijec.v4i2.110](https://doi.org/10.52015/nijec.v4i2.110)

Latency-Aware Edge–Cloud Perception Architecture for Latency-Sensitive Medical Metaverse Applications [Under Review]

SSRN Preprint • DOI: ssrn.com/abstract=6254021

- Proposes a hybrid edge–cloud inference framework using YOLOv8-tier detectors and a Latency-Aware Decision Module (LDM) for real-time medical XR perception — achieving 83.6% latency reduction vs. cloud-only deployment while maintaining 69% detection agreement.

Development and Validation of the XR Readiness Index (XRI): A Computational Framework for Assessing Extended Reality Adoption Preparedness in Medical Education — A Pakistan Case Study

Shah, A., & Khan, A. (2026). Zenodo Preprint • DOI: [10.5281/zenodo.20026564](https://doi.org/10.5281/zenodo.20026564)

- Introduces the XR Readiness Index (XRI), the first validated computational framework for quantitatively assessing institutional XR adoption preparedness in LMIC medical education — validated across 191 respondents from all seven provinces of Pakistan (mean XRI = 58.69/100; Cronbach’s α = 0.750–0.801; EFA confirmed five-factor structure, KMO = 0.865).

RESEARCH INTERESTS

- Real-time deep learning–based object detection on resource-constrained embedded and edge devices
- Lightweight neural network architectures and model optimization (quantization, pruning, knowledge distillation)
- Performance–efficiency trade-offs in deploying vision AI on low-power hardware platforms

RESEARCH & TECHNICAL PROJECTS

Real-Time License Plate Detection & Recognition System

Python • TensorFlow • OpenCV • SSD MobileNet V2 • EasyOCR • github.com/AwaisShah75/Real-Time-License-Plate-detection-System

- Fine-tuned SSD MobileNet V2 on 400+ annotated images for multi-format plate localisation across varying lighting conditions.
- Engineered a custom OCR filtering mechanism using 60% region-based thresholding to isolate text from background noise.
- Designed a UUID-based logging system exporting to CSV; optimised CPU inference for parallel TensorFlow + PyTorch execution.
- Deployed as a prototype for parking automation, traffic monitoring, and access-control systems.

Real-Time Sign Language Detection System

Python • TensorFlow • OpenCV • SSD MobileNet V2 • github.com/AwaisShah75/Real-Time-Sign-Language-Detector

- Built a custom dataset of 280+ images annotated in Pascal VOC XML format; fine-tuned SSD MobileNet V2 over 15,000 iterations.
- Achieved AP@50: 0.990 • AP@75: 0.708 • mAP (0.50–0.95): 0.608 • AR@100: 0.677.
- Designed a real-time inference pipeline with bounding box visualisation, confidence scoring, and low-latency processing.

TEACHING & OUTREACH

Computer Science Teacher — Brainiac International School & College — Aug 2024 – Present

- Improved measurable understanding for 53% of students using the 5E Model, interactive examples, and conceptual learning.

- Guided students in problem-solving, project-based learning, and introductory AI modules.

TECHNICAL SKILLS

Programming	Python
ML Frameworks	TensorFlow, Keras, PyTorch, OpenCV, EasyOCR, NumPy, Pandas
Tools	Git, VS Code, Jupyter Notebook, Google Colab, Matplotlib, LabelImg
Domains	Embedded AI, Edge Computing, Computer Vision, Deep Learning (CNNs)
Languages	Pashto (Native), Urdu (Native), English (Professional)

CERTIFICATIONS & ACHIEVEMENTS

- Python Programming — Kaggle (2025)
- Introduction to Machine Learning — Kaggle (2025)
- Guest Presenter, Annual IT Exhibition (2025) — ANPR system demonstration to GM of WAPDA; recognized for innovation and execution.